

1. Set Up Your Tools

- Before writing code, you need an environment to run it.
- Oracle Java Development Kit (JDK)
- Integrated Development Environment (IDE)

Visual Studio Code for Java development IDE

- Microsoft developed Visual Studio Code (VS Code) IDE.
- It is a fast, lightweight, and highly customizable code editor that has become an excellent option for Java programming. Seamlessly, work with other programming languages as well like Python and JavaScript.
- Alternatively, if you prefer not to install a JDK or use a local IDE, you can use online compilers like [Programiz](https://www.programiz.com/java-programming/online-compiler/) for quick programming. <https://www.programiz.com/java-programming/online-compiler/>

Getting Started Workflow

1. **Install the Software:** Use the unified "Coding Pack for Java" installer (available for Windows and macOS) to get **VS Code (IDE), the JDK, and crucial extensions simultaneously**. Download and install from: https://code.visualstudio.com/docs/java/java-tutorial#_setting-up-vs-code-for-java-development
2. **Generate a Project:** Press Ctrl+Shift+P (or Cmd+Shift+P on Mac) to open the Command Palette, type Java: Create Java Project, and choose No build tools for a clean starter setup.
3. **Write Code:** Build files inside the generated src folder using [integrated productivity shorthand snippets](#) like main or sysout.
4. **Execute:** Launch your app smoothly by clicking the built-in **Run** or **Debug** CodeLens buttons directly above your main class file.
5. **Getting started with VSCode** <https://code.visualstudio.com/docs/java/java-tutorial>

Follow the link to download and install Coding Pack for Java

1> Video link: <https://www.youtube.com/watch?v=tYYVeCB7Ef4>

Tutorial for using VS Code

<https://www.youtube.com/watch?v=bj6QSx22zXs>

What is IDE

An IDE, or Integrated Development Environment, is a software application that combines all the essential tools a developer needs to write, test, and debug code into a single program. Instead of using a separate text editor, compiler, and terminal, an IDE merges them into one interface to make programming faster and easier.

Core Components of an IDE

- **Source Code Editor:** A specialized text editor with features like auto-complete, syntax highlighting, and error checking.
- **Build Automation Tools:** Tools that automatically compile and build your raw code into an executable program.
- **Debugger:** A built-in diagnostic tool that lets you pause your program step-by-step to find and fix bugs.

IDE vs. Text Editor

- **Text Editors (e.g., Notepad++, basic VS Code):** Lightweight programs meant for typing raw text. They require you to manually install plug-ins or use external terminal commands to run your code.
- **IDEs (e.g., Eclipse, IntelliJ IDEA, Visual Studio):** Heavyweight applications that come completely pre-configured out of the box with compilers, linkers, and full execution environments.